

## Ch-2

# Lens: Lens refer to a piece of glass, that is shaped like a lentil seed, thick at middle & thinner from sides (This define is based on this chapter & this structure is of a convex lens). Lens are used for making microscopes & are very important for researches. Early microscope were not that powerful but led to the discovery of these organisms that an unaided eye can not see.

# Cell is the smallest structural & functional unit of life. Every living being is made up of cells. Some organisms are made up of one cell only & are called unicellular organisms & some organisms have more than one cell, they are known as multicellular organisms.

# Cell is the basic unit of life because cells can work independly & control their growth & energy by themselves. This is also the reason why unicellular organisms survive, because they can do every function.

## # Cells mainly have these three parts:

- Cell membrane - It ~~se~~ separates a cell from another cell. It is porous so it allows entry of substances & exit of waste.
- ~~Cell~~ Cytoplasm - It is present inside cell membrane & is a jelly-like liquid which has essential nutrients. Most life processes happen in it.
- Nucleus - Regulate all activities inside the cell like hormone & growth. Bacterias don't have this but they have a nucleoid.

## # Spl. parts

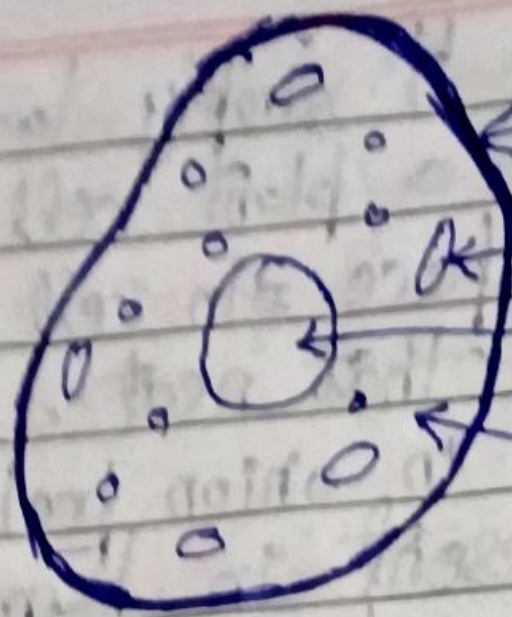
- Cell wall - Acts a covering for the whole cell. It provides ~~high~~ rigidity & support for standing upright. It is also porous.
- Plastids - Tiny rod-like structures in plant cells, Eg - Chloroplasts. In non-green plants in acts as an ~~store~~ storage box.

\* Cell wall is present in both plant cell & ~~fungi~~ fungi cells.

- Vacuole - It simply looks like an empty space in a plant cell, which takes up 90% of space in cell but it helps maintain that neat & arranged pattern of cell (Like in onion peel experiment). It's rarely present in ~~human~~ animal cells, but when it is, it is very small.

| # Plant Cells                               | Animal Cells                                 |
|---------------------------------------------|----------------------------------------------|
| They have cell walls                        | They don't have cell walls                   |
| They have plastids                          | They don't have plastids                     |
| They have 1 vacuole that is big & only one. | Vacuoles are smaller & there can be multiple |

Dia-1:



Cell membrane  
Mitochondria  
Nucleus  
Cytoplasm

Dia-2:



Nucleus

Mitochondria

Cell wall

Cell membrane

Vacuole

Cytoplasm

Plastid (Chloroplast)

Muscle cell

- Spindle shaped appearance
- Thin & flexible, works by contraction & relaxation to produce force.
- Eg - Heart, Digestive system

Nerve cell (Neuron)

- Long & branched appearance
- Transport messages & signals from brain to other parts of body or vice versa.

## # Levels

Cells → Tissue → Organ → Organ System → Organism

# Egg: It is a cell that can repeatedly divide to form a complete living being made up of many cells.

# Largest cell ever found is the yolk of an ostrich egg (ranging from 150 - 170 m)

# Some organisms that can not be seen with unaided eyes, are called micro-organism & or microbes. Eg - Fungi. They're usually observed with an microscope.

# Even air, water, soil ~~suspe~~ ~~set~~ suspension & stagnant water has microbes.

↳ Microbe commonly found in:  
Stagnant water: Amoeba (Protozoa), Paramecium (Protozoa), Algae

Soil suspension: Mould (Fungi), Bread Mould (Fungi), Algae, Bacteria

## # Microbes (Types)

Fungi   Protozoa   Bacteria   Algae  
Virus

## # USES OF MICROBES

- Used in Baking & Beer Industries (Yeast is the name of that microbe)
- Lactobacillus is commonly used to curdle the milk
- Microalgae produce more than 50% of Earth's oxygen. They are also used to make health supplements. Eg - Spirulina as a source of protein, Vitamin B<sub>12</sub>.
- Rhizobium is a bacteria which live in root nodules of legumes & does nitrogen fixation, ultimately increasing nitrogen in soil leading to better crop growth.

## # HARMFUL EFFECTS OF MICROBES

- They can cause disease that can be life threatening also. Eg - Typhoid, Malaria
- They can spoil things. Eg - Mould on Bread. That's why we use food preservation methods. Eg - Pickling (Adding a lot of salt & spices)

## # Scientists

- Robert Hooke: A scientist & a part of the ROYAL SOCIETY. Published a book called Micrographia, in which he used the word 'cell' for first time ever in science to describe the basic unit of life. He used an microscope to see beyond our eyes & drew detailed diagrams of what he observed.
- Antonie Van Leeuwenhoek: A dutch scientist known as the "Father of Microbiology" as he made more powerful microscopes & saw ~~bas~~ things like blood cells & bacteria & described his sighting.
- Anand Mohan Chakrabarty - He was an India scientist who got a patent for developing a spl. bacterium that can break down oil spills & help clean the environment.